

Geometric Intuition of Hilbert Spaces

Banach spaces admit a notion of size per elements. We can estimate for how this notion of size changes when adding or multiplying them (triangle inequality, Minkowski inequality, etc.). A key feature that is not present in these spaces is the granular information about how much two elements are similar. For example, how much is a vector a *linear* combination of a single series of functions with different scalar weights, rather than just the convergence of a series where we cannot isolate the “contribution” of one element over another. If we had this information, we could project onto basis elements and meaningfully compare elements. In the same vein, is there a way to measure the “defect” in the triangle inequality? That is, can we find a reliable functional $J(f, g)$ such that:

$$\|f + g\|_p^2 = \|f\|_p^2 + \|g\|_p^2 + J(f, g)$$

Such niceties are in fact not found in general Banach spaces due to algebraic blockers (or, interpreted geometrically, due to the non-flat geometry of most Banach spaces).

To see this, let us consider how to connect an element $f \in L^p$ to its natural functional (we shall justify natural shortly). First, what does it mean to go in the “same direction” in a function space? (For simplicity we’ll always work with L^p spaces¹). If $f, g \in L^p$, then we would say that $f + g$ implies moving in the direction of g along the “same line”; this matches our finite-dimensional intuition. Then $f + f = 2f$ implies moving in the same direction as f . Let us say that we want a functional that represents going in the direction of f , namely $\varphi_f \in (L^p)^*$ such that $\varphi_f(f) = \|f\|_p$. This is a natural choice: in finite dimensions we interpret functionals as measuring how much a vector projects onto a direction. As $(L^p)^* \cong L^q$, it is easy to see that this functional corresponds to $g \in L^q$ such that:

$$g(x) = |f(x)|^{p-1} \text{sgn}(f(x))$$

Note that $g \in L^q$ since:

$$|g|^q = |f|^{q(p-1)} = |f|^{qp-q} \stackrel{!}{=} |f|^p$$

where $\stackrel{!}{=}$ comes since $1/p + 1/q = 1$ iff $p + q = pq$ iff $p = q(p-1)$. Hence as $f \in L^p$

$$\int |g|^q = \int |f|^p < \infty$$

Thus, we have a map $f \mapsto \varphi_f$. Let us define the duality map:

$$J_p : L^p \rightarrow L^q \quad f \mapsto \varphi_f$$

This map is *not* linear:

$$J_p(2f) = |2f|^{p-1} \text{sgn}(2f) = 2^{p-1} |f|^{p-1} \text{sgn}(f) = 2^{p-1} J_p(f)$$

As $(L^q)^* \cong L^p$, we can interpret this as a map $J_p : L^p \rightarrow (L^p)^*$. This is only linear in the special case of $p = 2$, since:

$$J_2(nf) = |nf|^{2-1} \text{sgn}(nf) = n J_2(f)$$

¹the reasoning can extend to Fréchet by taking limit of Hilbert spaces, but a new approach is needed for metric spaces, called CAT(k) geometry, see [2] for more details

Why is this specific functional the “natural” choice? In functional analysis, the Hahn-Banach theorem guarantees that for any non-zero element f , there exists at least one functional φ of unit norm such that $\varphi(f) = \|f\|$. Geometrically, the kernel of this functional defines a closed hyperplane that touches (supports) the ball of radius $\|f\|$ exactly at the point f .

In rougher spaces like L^1 or L^∞ , the unit balls have sharp corners and flat edges. At a corner, you can balance infinitely many different flat sheets (hyperplanes) on the tip; essentially, the “direction” of the functional is ambiguous. However, for $1 < p < \infty$, the L^p spaces are smooth and strictly convex. Note that a norm is smooth if the Gâteaux derivative for any two x, y away from the origin exists:

$$\lim_{t \rightarrow 0} \frac{\|x + ty\| - \|x\|}{t} \text{ exists.}$$

For L^p where $1 < p < \infty$, the unit ball is smooth. Consequently, at every point f on the surface, there is exactly one unique tangent plane. The duality map $J_p(f)$ we derived above is simply the unique algebraic description of this tangent plane. It is the only functional that aligns perfectly with the geometry of the surface at that specific point.

This can be thought of geometrically: when $p \neq 2$, as we move further from the origin, the ‘strength’ or sensitivity of the geometry grows non-linearly. This effectively warps the space so that large vectors interact differently than small ones (in terms of physics intuition, the “cost” of moving away from the origin depends on how far out you are).

Let us connect this using actual geometry. Consider the function that defines the geometry of the L^p unit ball:

$$F(x) = \frac{1}{p} \|x\|_p^p = \frac{1}{p} \sum |x_i|^p$$

(where the ℓ^p norm is used for visualization, but the logic holds for integrals). From multivariable calculus, we know the unit sphere is the Level Set $F(x) = \frac{1}{p}$, and the gradient $\nabla F(x)$ is the vector strictly perpendicular (normal) to the level set at point x . Calculating the gradient:

$$\frac{\partial}{\partial x_i} \left(\frac{1}{p} |x_i|^p \right) = |x_i|^{p-1} \text{sgn}(x_i)$$

So, the gradient vector is:

$$\nabla F(x) = (|x_1|^{p-1} \text{sgn}(x_1), \dots, |x_n|^{p-1} \text{sgn}(x_n))$$

Notice that this is exactly our duality map:

$$\nabla F(x) = J_p(x)$$

Thus, $J_p(x)$ is the right candidate to represent the geometry of L^p ².

Let us now understand how this relates the inner product, in particular consider how we recover the inner product in Euclidean space. If we define the energy of a vector as $E(x) = \frac{1}{2} \|x\|_2^2$, then the directional derivative of this energy at x in the direction of a perturbation y is:

$$\left. \frac{d}{dt} E(x + ty) \right|_{t=0} = \left. \frac{d}{dt} \frac{1}{2} \langle x + ty, x + ty \rangle \right|_{t=0} = \langle x, y \rangle$$

Thus, the inner product $\langle x, y \rangle$ is not just an algebraic artifact; it physically represents the *sensitivity* of the norm of x to a perturbation by y .

²If $p = 1$, notice that $J_1(x) = \text{sgn}(x)$ has infinitely many possibilities at zero, mirroring the fact that the L^1 unit ball is a jagged diamond with corners having infinitely many tangents.

In L^p spaces, we lack a global inner product, but we can still ask this fundamental question: how does the p -energy $F(f) = \frac{1}{p}\|f\|_p^p$ change if we perturb f by a function g ? Using the gradient we just calculated, the first-order variation is³:

$$dF(f; g) = \langle \nabla F(f), g \rangle_{\text{dual}} = \int \nabla F(f) \cdot g$$

Substituting $\nabla F = J_p(f)$, this variation becomes exactly $\int J_p(f)g$. Thus, we can define a pairing:

$$\langle f, g \rangle_p = \int J_p(f)g$$

Note that it is important to go through $J_p(f)$ since $\int fg$ need not be finite in general. However, by Hölder's Inequality:

$$\int |J_p(f)g| < \infty$$

As noted, this is *not* an inner product, primarily because it is not symmetric! This hints that the geometry of L^p is *not* flat for $p \neq 2$. If however $p = 2$, then $J_p(x)$ maps identically to itself and we recover the symmetric form $\int fg$.

This lack of symmetry in the pairing $\langle f, g \rangle_p$ is directly linked to the fact that L^2 is the unique L^p space that is *rotationally invariant*. Geometrically, rotational invariance means that the properties of a vector depend only on its length, not on its alignment with the coordinate axes. Mathematically, this requires that the duality map commutes with rotations. If R is a rotation operator (an isometry), we expect the “normal vector” of the rotated point to be the rotated “normal vector” of the original point:

$$J_p(Rf) \stackrel{?}{=} R(J_p(f))$$

In L^2 , since J_2 is linear (essentially the identity), this holds trivially: $J_2(Rf) = Rf = R(J_2(f))$. However, for $p \neq 2$, the map $f \mapsto |f|^{p-1}\text{sgn}(f)$ applies a non-linear distortion component-wise. Rotating the vector mixes the components, and applying the power law *after* mixing is not the same as applying it *before*.

To see this concretely, consider ℓ^3 ($p = 3$) in $\mathbb{R}^2$ and let $f = (1, 0)$. If we first compute the functional and then rotate by $\pi/4$ (45°), we get:

$$R(J_3(f)) = R(1^2, 0^2) = R(1, 0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

However, if we rotate the vector first, the components become $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$. Applying the duality map J_3 (squaring components) yields:

$$J_3(Rf) = J_3\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Since $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \neq (\frac{1}{2}, \frac{1}{2})$, we have $J_p(Rf) \neq R(J_p(f))$. This confirms that for $p \neq 2$, the geometry is *anisotropic*: the shape of the unit ball looks different depending on the angle from which you view it.

This distinction can be understood most rigorously through the lens of Riemannian geometry. While the pairing $\langle \cdot, \cdot \rangle_p$ corresponds to the first derivative (gradient) of the energy, the geometry of the space is determined by how this gradient *changes*—the second derivative, which acts as the metric tensor. In this framework, the metric tensor $g_{ij}(x)$ at a point x dictates how to measure the length

³The dF here can be thought of as the Gâteaux derivative, commonly defined for functions between locally convex topological vector spaces

of tangent vectors. In our context, this metric is generated by the Hessian of the energy function $E(x) = \frac{1}{p}\|x\|_p^p$. For L^2 , this Hessian is proportional to the identity operator everywhere; the metric is *constant* (flat). However, for L^p ($p \neq 2$), the second derivative depends on position x , specifically scaling as $|x|^{p-2}$. This implies that the “metric” changes as we move around the space.

To visualize the practical consequences of this non-flat geometry, it is helpful to consider the “shape” of the unit ball in infinite dimensions. In finite dimensions, all norms are equivalent, meaning the unit ball of one norm can always be fit inside a scaled unit ball of another. For example, in $\mathbb{R}^n$, the ℓ^1 diamond fits inside the ℓ^∞ cube, and the cube fits inside a scaled diamond. The ratio of their sizes is bounded by the dimension n . This means that a “spike” in the ℓ^1 ball has a finite maximum length relative to the ℓ^∞ norm; you cannot travel infinitely far down a spike without eventually hitting the origin in every other norm. Topologically, this guarantees that convergence in one norm implies convergence in all others.⁴

In infinite dimensions, this equivalence breaks because the dimension n — and thus the scaling factor between norms — goes to infinity. The curvature of the norm now dictates how “energy” is allowed to concentrate without bound. The L^∞ unit ball acts like a strict box: it bounds the height of a function everywhere, preventing energy concentration but allowing mass to spread out infinitely. Conversely, the L^1 unit ball allows for infinitely long thin spikes: it allows functions to become arbitrarily tall as long as they are thin enough (concentration of measure), effectively extending “spikes” out to infinity relative to the L^∞ norm.

This geometric disparity explains why convergence depends on the norm. Convergence $f_n \rightarrow 0$ is not about movement, but about entering arbitrarily small ϵ -balls. Consider the sequence $f_n(x) = x^n$ on $[0, 1]$. In the L^1 geometry, this sequence travels down one of the “spikes” of the unit ball: the functions get thinner and thinner, their area (and thus distance to the origin) shrinking to zero despite their height remaining constant. In the L^∞ geometry, however, these functions are trapped on the surface of the unit sphere (height 1); they cannot enter the ϵ -ball because the L^∞ “box” does not extend down the “spikes” of concentration.

The Hilbert space L^2 sits perfectly in the middle. Its unit ball is a round sphere, balancing height and width. If L^1 can be thought of as having no penalties to concentrate energy, and L^∞ has no penalty in spreading mass, L^2 creates a geometry that penalizes both extreme concentration (spikes) and extreme spread, providing the only setting where energy behaves isotropically (again, visualized by the fact the unit ball has rotational invariance). Thus, while L^2 is a flat Euclidean plane, general Banach spaces behave like curved manifolds where the very definition of angle and length is position-dependent. For further geometric discussion of Banach spaces, see *Geometry of Banach Spaces* by Diestel [1].

With this understanding of the position-dependent metric, we can clarify the notion of orthogonality. In Euclidean geometry, orthogonality is symmetric ($x \perp y \iff y \perp x$) and additive. In general Banach spaces, without a global bilinear form, we must rely on *Birkhoff-James orthogonality*: we say $x \perp y$ if moving in the direction of y does not decrease the length of x ($\|x + \lambda y\| \geq \|x\|$). Geometrically, this means the line passing through x in the direction of y is tangent to the unit sphere at x .

Analytically, utilizing the functional J_p , this condition is equivalent to requiring that y be annihilated by the functional generated by x :

$$\langle J_p(x), y \rangle = 0$$

This reveals the fundamental flaw in non-Hilbert geometry. In L^2 , because J_2 is linear, the condition is symmetric. However, in L^p ($p \neq 2$), the non-linearity of J_p breaks reciprocity. The set of vectors orthogonal to x (kernel of $J_p(x)$) does not align symmetrically with the set of vectors orthogonal to y . Consequently, it is common to have $x \perp y$ but $y \not\perp x$. This lack of symmetry illustrates why

⁴in particular, they define the same *topology* even if the norms define different *geometries*

Hilbert spaces are the unique setting where Euclidean geometric intuition remains valid.

The unique geometry of Hilbert spaces—specifically self-duality, isotropy, and the inner product—unlocks powerful results unavailable in general Banach spaces. The following are key consequences we will explore:

1. *The Projection Theorem (Best Approximation)*: In general Banach spaces, finding the element in a subspace M closest to a point x is difficult (the closest point might not be unique, e.g., the flat edge of L^1). In Hilbert spaces, the strict convexity and “roundness” of the unit ball guarantee that for any closed subspace M , there exists a *unique* element $P_M x$ minimizing the distance $\|x - m\|$. Furthermore, the error vector $x - P_M x$ is orthogonal to the subspace.
2. *The Riesz Representation Theorem (Self-Duality)*: The duality map $J : H \rightarrow H^*$ is linear and bijective. This means we can completely identify linear functionals with vectors. Every continuous linear functional φ acts exactly like an inner product: $\varphi(x) = \langle x, y \rangle$ for a unique y . This collapses the study of H^* to the study of H .
3. *Orthonormal Bases and Fourier Series*: In L^p , we may have a Schauder basis⁵, but determining the coefficients is non-trivial. In Hilbert spaces, orthogonality allows us to decouple coefficients. This is key for the Fourier transform, where any element is reconstructed by projection: $x = \sum \langle x, e_n \rangle e_n$. Additionally, the *Parseval Identity* allows us to measure size by summing squared coefficients: $\|x\|^2 = \sum |\langle x, e_n \rangle|^2$.
4. *Existence of Adjoints*: In Banach spaces, the adjoint of $T : X \rightarrow Y$ maps $Y^* \rightarrow X^*$. Because Hilbert spaces are self-dual, we can pull this operator back to the original space. Thus, for every bounded operator T on a Hilbert space, there is a unique adjoint T^* on the *same space* satisfying $\langle Tx, y \rangle = \langle x, T^*y \rangle$. This is foundational for spectral theory.

In summary, while Banach spaces provide the necessary topological framework (completeness), Hilbert spaces add the rigid geometric structure required to generalize linear algebra to infinite dimensions.

⁵A sequence $\{b_n\}$ where every $v \in V$ has a unique expansion $v = \sum \alpha_n b_n$